Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Mass	Power Demand	Processing	Radiation Durability	Reliability/TRL	Weight	Normalized
Mass		0.25	0.25	0.25	0.25	1	0.10
Power Demand	0.75		0.25	0.25	0.25	1.5	0.15
Processing Throughput	0.75	0.75		0.25	0.25	2	0.20
Radiation Durability	0.75	0.75	0.75		0.75	3	0.30
Reliability/TRL	0.75	0.75	0.75	0.25		2.5	0.25
					Total	10	

	Mass	Power Demand	Processing	Radiation Durability	Reliability/TRL
5	< 3 kg total	< 15 W	> 1000 MIPS; >50%	> 200 krad TID, latch-up	TRL 9; flown at Jupiter-
4	3 - 4.5 kg	15 - 25 W	400 - 1000 MIPS	100 - 200 krad, SEL	TRL 8; flown in deep
3	4.5 - 6 kg	25 - 40 W	200 - 400 MIPS	50 - 100 krad; needs	TRL 6-7; qualified but
2	6 - 8 kg	40 - 60 W	100 - 200 MIPS	20 - 50 krad; current-	TRL 5; prototype,
1	> 8 kg	> 60 W	< 100 MIPS; no	< 20 krad or SEL-	TRL < 5

Decision Criteria		Analysis of the onboard flight computer and mass storage architecture					
		RAD750-class SBC + 1 GB rad-hard NAND storage board		RAD5545 quad-core PowerPC + 4 GB storage		COTS ARM SoC (Zynq-class) with TMR voting and software EDAC	
Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Mass	0.1	4	0.4	3	0.3	5	0.5
Power Demand	0.15	4	0.6	3	0.45	5	0.75
Processing Throughput	0.2	3	0.6	5	1	5	1
Radiation Durability	0.3	5	1.5	4	1.2	2	0.6
Reliability/TRL	0.25	5	1.25	3	0.75	2	0.5
Total Score		4.35		3.70		3.35	

The RAD750-class board scores 4.35 against 3.70 for the RAD5545 and 3.35 for a COTS ARM SoC with triple-modular-redundant voting. Radiation durability and flight heritage carry 55% of the weight between them, and that is what decides this. RAD750 parts are latch-up immune by process and qualified past 100 krad, which is the number REQ.01.02 already fixed, and the part flew to Jupiter on Juno and is flying there again on Europa Clipper. Nothing else in the trade has seen this environment. Throughput is where it gives ground: roughly 266 MIPS at 200 MHz against a peak load near 170 MIPS during Traverse, when visual odometry, hazard assessment, and the navigation EKF all run inside one arc cycle. That leaves 36% margin against the 30% floor in REQ.12.01, so the answer is adequate rather than comfortable, and it is the first term to watch if the VO pipeline grows. The RAD5545 buys throughput the mission does not need. Peak generation across the whole rover is 29.7 kbps and the buffer never climbs past 9% of capacity on a nominal sol, so there is no processing or storage problem worth 5.5 kg and a lower TRL. The COTS SoC is the tempting one and the wrong one. It wins mass, power, and throughput outright, then loses on the only axis that ends missions here: at 20 to 50 krad it is consumed inside the 30-sol surface phase even behind the vault, and TMR voting protects a running computation against upsets while doing nothing about total dose. With a 44 min one-way delay there is no ground recovery for a part that has simply worn out. Selected: RAD750-class SBC with a 1 GB rad-hard NAND storage board.